

Eunhan Ka

POSTDOCTORAL RESEARCHER · LYLES SCHOOL OF CIVIL AND CONSTRUCTION ENGINEERING, PURDUE UNIVERSITY
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Education

Purdue University

United States

PH.D., LYLES SCHOOL OF CIVIL AND CONSTRUCTION ENGINEERING

Dec. 2025

- **Dissertation: Physics-Informed Neural Networks for Secure Connected and Autonomous Traffic Modeling**
- Committee: Satish V. Ukkusuri (Chair), Ludovic Leclercq, Yiheng Feng, Z. Berkay Celik

Seoul National University

South Korea

M.S., DEPARTMENT OF CIVIL AND ENVIRONMENTAL ENGINEERING

Feb. 2018

Seoul National University

South Korea

B.S., DEPARTMENT OF CIVIL AND ENVIRONMENTAL ENGINEERING

Feb. 2016

- Completed the ABEEK-accredited Civil and Environmental Engineering program (Engineering Accreditation Commission; Washington Accord).

Research Interests

Artificial intelligence (AI), optimization, and network modeling for transportation systems, emphasizing state estimation from incomplete observations and how information reliability shapes traveler behavior and network dynamics. I use traffic flow theory, network structure, and behavioral models to support transportation operations.

- **Themes:** Reliable Network Monitoring and State Estimation; Transportation Network Operations and Resilience; Traveler Behavior and Information Reliability; Connected and Automated Mobility.
- **Methods:** Physics-Informed and Graph Learning; Generative Models; Traffic-Flow and Network Modeling; Behavioral Analysis; Traffic Assignment, Optimization, and Simulation; Spatiotemporal Data Analytics.

Selected Publications & Awards

Selected Publications

1. **Ka, E.**, Leclercq, L., and Ukkusuri, S. V. (2027). Observation-aligned two-stage domain decomposition for physics-informed traffic state estimation with sparse fixed sensors. *Transportation Research Part C: Emerging Technologies*, 194, 105995. Published online September 11, 2026.
2. **Ka, E.**, Xue, J., Leclercq, L., and Ukkusuri, S. V. (2024). A Physics-Informed Machine Learning for Generalized Bathtub Model in Large-Scale Urban Networks. *Transportation Research Part C: Emerging Technologies*, 164, 104661.
3. Xue, J., **Ka, E.**, Feng, Y., and Ukkusuri, S. V. (2024). Network Macroscopic Fundamental Diagram-Informed Graph Learning for Traffic State Imputation. *Transportation Research Part B: Methodological*, 189, 102996.
4. **Ka, E.** and Ukkusuri, S. V. (2026). Resilience of Traffic Networks to Route Guidance Attacks: The Role of Driver Behaviour Heterogeneity. *Transportmetrica B: Transport Dynamics*, 14, 1.
5. **Ka, E.** and Ukkusuri, S. V. (2025). Route Guidance Attacks in Cyber Transportation Networks: A User-Centered Study of Behavioral Sensitivity. *Transportation Research Part F: Traffic Psychology and Behaviour*, 115, 103354.

Selected Awards & Research Support

1. **2025 ITE Great Lakes District Student Paper Award** (First Place in regional student paper competition), *Great Lakes District of ITE (GLITE)*. Jun. 2025
2. **Honorable Mention, 2025 Clifford Spiegelman Student Paper Competition**, *The Transportation Statistics Interest Group of the American Statistical Association*. Jan. 2025
3. **ChatGPT for Academic Researchers Program, Approved Participant**, *OpenAI*. Aug. 2026
4. **Google Cloud Research Credits Program, Credit Recipient**, *Google Cloud*. Aug. 2025 & Jun. 2026

Research Experience

Postdoctoral Researcher

Jan. 2026 - Present

PURDUE UNIVERSITY, UNITED STATES

- Develop **physics-informed methods for traffic-state reconstruction** from sparse fixed-sensor observations, including observation-aligned domain decomposition for offline speed-field reconstruction.
- Model **route-guidance misinformation and heterogeneous traveler responses** to evaluate modeled route choices, travel times, and aggregate network dynamics.
- Develop **day-to-day traffic-network models with endogenous trust and adaptive reliance on guidance**; examine network performance and recovery in ongoing work under review.
- Contribute to **research proposal development** for the U.S. Department of Transportation (USDOT) University Transportation Centers (UTC) Program, Indiana Department of Transportation (INDOT), and NSF Foundational Research in Cyber-Physical Systems (CPS-FR).

Graduate Research Assistant

Aug. 2020 - Dec. 2025

PURDUE UNIVERSITY, UNITED STATES

- Developed physics-informed and graph-based methods for **large-scale traffic-state estimation under sparse or disrupted data** and analyzed **route-choice responses and travel-time impacts of unreliable guidance**. (*Sponsor: USDOT UTC Program, National Center for Transportation Cybersecurity and Resiliency (TraCR)*)
- Conducted a statewide workforce needs assessment to identify skill gaps and designed training strategies to improve engagement and performance (presented at **Purdue Road School, Mar. 2025**). (*Sponsor: INDOT*)
- Analyzed network-level impacts of a movable median barrier on the Alex Fraser Bridge using mobile location data (*Journal of Transportation Engineering, Part A*, 2022; **Editor's Choice**). (*Sponsor: Lindsay Corporation*)
- Led proposal development and served as the **primary technical lead throughout project execution**, coordinating analytical work, project milestones, and deliverable preparation under faculty Principal Investigator Satish V. Ukkusuri. **Selected projects and project periods:**
 1. Attracting and Retaining the Transportation Workforce and At Risk Targeted Areas (*Sponsor: INDOT, total project award: \$225,000, period: Oct. 2024–Sep. 2026*).
 2. A Multi-Resolution Simulation Platform for Transportation System Security Testing and Evaluation (*Sponsor: USDOT UTC-TraCR, total project budget: \$340,000 including cost share, period: Jan.–Dec. 2024*).
 3. Training Gap Analysis for INDOT Workforce (*Sponsor: INDOT, total project award: \$213,500, period: Oct. 2023–Oct. 2025*).

Researcher

Mar. 2018 - Aug. 2020

SEOUL NATIONAL UNIVERSITY, SOUTH KOREA

- Designed cluster-based routes for demand-responsive transport using multi-capacity vehicles to serve passengers with disabilities (presented at **99th TRBAM**). (*Sponsor: T-Money Welfare Foundation*)
- Modeled lane-changing on freeways and gap-acceptance behavior at roundabouts using real-world data and virtual reality simulations (published in **Journal of Advanced Transportation, 2018**).

Graduate Research Assistant

Mar. 2016 - Feb. 2018

SEOUL NATIONAL UNIVERSITY, SOUTH KOREA

- Designed a real-time vehicle relocation strategy and an event-based simulation for one-way car-sharing services (presented at **multiple conferences, 2016-2018**). (*Sponsor: National Research Foundation of Korea*)
- Developed a methodology to evaluate driver behavior using surrogate safety measures (published in **Journal of Advanced Transportation, 2020**). (*Sponsor: Ministry of Land, Infrastructure and Transport, South Korea*)

Teaching & Mentoring Experience

Research Mentor

Fall 2024 - Present

LYLES SCHOOL OF CIVIL AND CONSTRUCTION ENGINEERING, PURDUE UNIVERSITY

- **Graduate research mentoring (Jun. 2026–Present):** Mentor two Ph.D. students through weekly research and manuscript feedback; each is first author on a transportation systems paper submitted to the 2027 Transportation Research Board Annual Meeting, with me as a coauthor.
- **Undergraduate research mentoring (Fall 2024–Spring 2025):** Mentored two undergraduate researchers weekly in transportation cybersecurity; their work resulted in a poster presentation at the 2025 Center for Education and Research in Information Assurance and Security (CERIAS) Cybersecurity Symposium.

Instructor

Fall 2026 - Present

LYLES SCHOOL OF CIVIL AND CONSTRUCTION ENGINEERING, PURDUE UNIVERSITY

- CCE 56900: Smart Logistics (Graduate)

Fall 2026

Teaching Assistant

Spring 2021 - Fall 2024

LYLES SCHOOL OF CIVIL AND CONSTRUCTION ENGINEERING, PURDUE UNIVERSITY

- CCE 66100: Algorithms in Transportation (*Graduate; co-developed new course curriculum*) Fall 2024
 - CCE 56601: Network Models for Connected and Autonomous Vehicles (Graduate) Fall 2021 - Fall 2023
 - CCE 56900: Smart Logistics (Graduate) Fall 2021, Fall 2024
 - CCE 56400: Data Science for Smart Cities (Graduate) Spring 2021, Fall 2023
- Developed online course materials for edX.

Lecturer

Fall 2019

DEPT. OF CIVIL AND ENVIRONMENTAL ENGINEERING, SEOUL NATIONAL UNIVERSITY

- Introduction to Civil and Environmental Engineering (Undergraduate)

Teaching Assistant

Fall 2016 - Fall 2017

DEPT. OF CIVIL AND ENVIRONMENTAL ENGINEERING, SEOUL NATIONAL UNIVERSITY

- Sustainable Transportation Systems (Graduate) Fall 2017
- Advanced Transportation Operation (Graduate) Spring 2017
- Leadership for Civil Engineers (Undergraduate) Spring 2017
- Introduction to Transportation Engineering (Undergraduate) Fall 2016

Publications

MANUSCRIPTS UNDER REVIEW AND REVISION

1. **Ka, E.**, Yeke, D., Celik, Z. B., and Ukkusuri, S. V. (Under Revision). Fake Rush Hour: Exploring Bounded Rational Route Guidance Attacks on Urban Navigation Systems. *ACM Journal on Autonomous Transportation Systems*.
2. **Ka, E.**, Mittal, S., Ali, A., Liu, L., and Ukkusuri, S. V. (Under Review). Worker-perceived regulatory calibration and workforce-impact indicators in U.S. transportation cybersecurity: Exploratory survey evidence. *Transport Policy*.
3. **Ka, E.** and Ukkusuri, S. V. (Under Review). Day-to-Day Traffic Network Modeling under Route-Guidance Misinformation: Endogenous Trust and Resilience in CAV Environments. *IEEE Transactions on Intelligent Transportation Systems*.

REFEREED JOURNAL PUBLICATIONS

1. **Ka, E.**, Leclercq, L., and Ukkusuri, S. V. (2027). Observation-Aligned Two-Stage Domain Decomposition for Physics-Informed Traffic State Estimation with Sparse Fixed Sensors. *Transportation Research Part C: Emerging Technologies*, 194, 105995. Published online September 11, 2026.
2. **Ka, E.**, Mittal, S., and Ukkusuri, S. V. (2026). Career-Stage-Specific Training Priorities for State Department of Transportation Workforce Development: An Integrated Regression and Importance-Performance Analysis. *Transportation Research Record: Journal of the Transportation Research Board*.
3. **Ka, E.** and Ukkusuri, S. V. (2026). Resilience of Traffic Networks to Route Guidance Attacks: The Role of Driver Behaviour Heterogeneity. *Transportmetrica B: Transport Dynamics*, 14, 1.
4. Ali, A., Liu, L., Ojeme, B., Anadozie, C., Agyemang, F., Ukkusuri, S. V., **Ka, E.**, and Mittal, S. (2026). Strengthening the Transportation Cybersecurity Workforce: A Mixed-Methods Analysis of Workforce Development, Organizational Preparedness and Cross-Sector Collaboration. *Future Transportation*, 6(4), 140.
5. **Ka, E.** and Ukkusuri, S. V. (2025). Route Guidance Attacks in Cyber Transportation Networks: A User-Centered Study of Behavioral Sensitivity. *Transportation Research Part F: Traffic Psychology and Behaviour*, 115, 103354.
6. Ukkusuri, S. V.*, Hamim, O. F.*, Lei, Z.*, **Ka, E.***, Salek, M. S., Chowdhury, M., Amini, M. H., Cardenas, A., and Thuraisingham B. (2025). Cybersecurity for Next-Generation Road Transportation: A Review. *ACM Journal on Autonomous Transportation Systems*. (*Contributed equally to this research).
7. **Ka, E.**, Xue, J., Leclercq, L., and Ukkusuri, S. V. (2024). A Physics-Informed Machine Learning for Generalized Bathtub Model in Large-Scale Urban Networks. *Transportation Research Part C: Emerging Technologies*, 164, 104661.

8. Xue, J., **Ka, E.**, Feng, Y., and Ukkusuri, S. V. (2024). Network Macroscopic Fundamental Diagram-Informed Graph Learning for Traffic State Imputation. *Transportation Research Part B: Methodological*, 189, 102996.
9. **Ka, E.**, Sharma, S., and Ukkusuri, S. V. (2022). Leveraging Location-Based Data for Assessing Network-Level Traffic Impact of Lane Management: A Case Study of Alex Fraser Bridge. *Journal of Transportation Engineering, Part A: Systems*, 148(12), 04022105. **[2022 Editor's Choice Collections]**
10. **Ka, E.**, Kim, D. G., Hong, J., and Lee, C. (2020). Implementing Surrogate Safety Measures in Driving Simulator and Evaluating the Safety Effects of Simulator-Based Training on Risky Driving Behaviors. *Journal of Advanced Transportation*, 2020(1), 7525721.
11. Lee, D., Hwang, S., **Ka, E.**, and Lee, C. (2018). Evaluation of the Rain Effects on Gap Acceptance Behavior at Roundabouts by a Logit Model. *Journal of Advanced Transportation*, 2018(1), 2726732.

CONFERENCE PAPERS AND PRESENTATIONS

1. **Ka, E.** and Ukkusuri, S. V. (2027). Local Identification of Origin-Destination Demand in Joint Demand-Cost Calibration from Partial Link Counts in Retained-Route Wardrop Models. *106th Annual Meeting of the Transportation Research Board*, Washington, D.C., United States.
2. **Ka, E.**, Li, H., and Ukkusuri, S. V. (2027). Finite-Grid Used-Path Regret Certificates for Restricted Static Traffic Assignment. *106th Annual Meeting of the Transportation Research Board*, Washington, D.C., United States.
3. Yang, Z., **Ka, E.**, Cai, Z., and Ukkusuri, S. V. (2027). Context-Matched Scenario Families with Controlled Criticality for Autonomous Driving Testing. *106th Annual Meeting of the Transportation Research Board*, Washington, D.C., United States.
4. **Ka, E.**, Yeke, D., Celik, Z. B., and Ukkusuri, S. V. (2026). MIRAGE: Detecting Fake Emergency Electronic Brake Light Attacks in V2X Networks via Event-Gated Behavioral Analysis. *4th USENIX Symposium on Vehicle Security and Privacy (VehicleSec'26)*, Baltimore, MD, United States.
5. Ortiz Barbosa, D., Abdisho, E. B., Chang, O., Burbano, L., Lei, Z., **Ka, E.**, Ukkusuri, S. V., Gilpin, L. H., and Cardenas, A. A. (2026). Beyond Attacker-Defender Ratios in Counter-Swarm Defense. *Dynamic Data Driven Applications Systems (DDDAS) 6th International Conference*, New Brunswick, NJ, United States.
6. **Ka, E.** and Ukkusuri, S. V. (2025). Analytical Framework for Network-Level Traffic Flow under Route Guidance Attacks: An Extension of the Generalized Bathtub Model. *104th Annual Meeting of the Transportation Research Board*, Washington, D.C., United States.
7. **Ka, E.** and Ukkusuri, S. V. (2025). Comparative Analysis of Sampling Methods in Physics-Informed Neural Networks for Traffic State Estimation in Large-Scale Road Networks. *104th Annual Meeting of the Transportation Research Board*, Washington, D.C., United States.
8. **Ka, E.**, and Ukkusuri, S. V. (2024). Impact of Cyber Attacks on Traffic State Estimation for Connected and Autonomous Vehicles Systems. *The Inaugural USDOT Future of Transportation (FoT) Summit*, Washington, D.C., United States.
9. Xue, J., **Ka, E.**, and Ukkusuri, S. V. (2024). Network Macroscopic Fundamental Diagram-Informed Graph Learning for Traffic State Imputation. *ISTTT25: 25th International Symposium on Transportation and Traffic Theory*, Ann Arbor, MI, United States.
10. **Ka, E.**, Xue, J., and Ukkusuri, S. V. (2024). PIDL-PedFlow: A Physics-Informed Deep Learning Approach for Macroscopic Continuum Pedestrian Flow Modelling. *103rd Annual Meeting of the Transportation Research Board*, Washington, D.C., United States.
11. Xue, J., **Ka, E.**, Mondal, W. U., and Ukkusuri, S. V. (2024). Generating Network-Level Dynamic Traffic Equations Using Symbolic Regression. *103rd Annual Meeting of the Transportation Research Board*, Washington, D.C., United States.
12. Sharma, S., and **Ka, E.** (2024). Leveraging Location-Based Data for Assessing Network Level Traffic Impact of Lane Management: A Case Study of Alex Fraser Bridge. *103rd Annual Meeting of the Transportation Research Board*, Washington, D.C., United States.
13. **Ka, E.**, and Ukkusuri, S. V. (2023). Dynamic Routing Games for Connected and Autonomous Vehicles with Traffic Congestion: A Mean Field Game Approach. *2023 INFORMS Annual Meeting*, Phoenix, AZ, United States.
14. **Ka, E.**, Ka, D., Jung, Y., and Lee, C. (2020). A Cluster-Based Route Design of Multi-Capacity Vehicle in Large-Scale Demand Responsive Transport Service for the Disabled. *99th Annual Meeting of the Transportation Research Board*, Washington, D.C., United States.

15. **Ka, E.**, Hong, D., Na, Y., and Lee, C. (2018). Analysis of Status in DRT Service for the Disabled in Seoul and Comparison Domestic and Foreign Cases. *International Conference for Road Engineers*, Jeju, South Korea.
16. Hong, D., **Ka, E.**, Ha, S., and Lee, C. (2018). Selection of Appropriate Hyperparameter for Waiting Time Prediction Model for Demand Responsive Transport for the Disabled in Seoul Using Long Short-Term Memory (LSTM) Network. *78th Korean Society of Transportation Conference*, Wonju, South Korea. **[Outstanding Paper Award]**.
17. **Ka, E.**, Kim, S., Hong, J., and Lee, C. (2017). A Preliminary Study of Comparison with Lane Changing Model Parameters in Merging Area between Normal and Raining Conditions. *12th International Conference of Eastern Asia's Society for Transportation Studies (EASTS)*, Ho Chi Minh, Vietnam.
18. **Ka, E.**, Kim, S., Woo, D., and Lee, C. (2016). An Estimation of Critical Gap for Gap Acceptance Model Applied to Lane Change of Surrounding Vehicles in Driving Simulator. *2016 Fall Korea Institute of ITS Conference*, Jeju, South Korea.
19. Lee, S., Lee, H., Lee, J., **Ka, E.**, and Lee, C. (2016). Analysis of Traffic Flow Impacts of Highway Traffic Accidents Using Survival Analysis. *74th Korean Society of Transportation Conference*, Jeju, South Korea. **[Outstanding Paper Award]**.

BOOK CHAPTERS AND TECHNICAL REPORTS

1. Dunlop, S. R., Ukkusuri, S. V., Chen, Y., Thakkar, D. J., **Ka, E.**, Mittal, S., Hu, X., Wang, Z., Khanna, I., and Tammineni, R. (2026). *INDOT Training Gap Analysis, Developing a Training Program for INDOT Workforce Development (New, Front-Line, Supervisors) and Smart Tool for Effective Planning for Workforce* (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2026/12). Purdue University.
2. **Ka, E.** and Ukkusuri, S. V. (2026). Misdirected Guidance and Urban Traffic Network Resilience. In *Advances in Transportation Cybersecurity and Resiliency*. World Scientific.
3. Hamim, O. F., Lei, Z., **Ka, E.**, and Ukkusuri, S. V. (2026). Understanding Cyber Threats in Cyber-Physical Mobility Systems. In *Advances in Transportation Cybersecurity and Resiliency*. World Scientific.
4. Verma, R., Luo, H., Deodhar, S., **Ka, E.**, Chahine, R., Natu, P., Malhotra, H., Polisetty, V., Thakkar, D. J., Ukkusuri, S. V., Cai, H., Dunlop, S. R., Iyer, A. V., and Gkritza, K. (2023). *Forecasting Shifts in Hoosiers' Travel Demand and Behavior* (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2023/28). Purdue University.

PATENTS AND INTELLECTUAL PROPERTY

1. Ukkusuri, S. V. and **Ka, E.** (2026). System and Method for Trusted Route Guidance with Anomaly Detection. *U.S. Provisional Patent Application No. 63/989,950, filed Feb. 24, 2026. Purdue Research Foundation. Patent pending.*

Awards, Fellowships & Research Support

Postdoctoral Supplemental Travel Award	Oct. 2026
Office of the Vice Provost for Graduate Students and Postdoctoral Scholars, Purdue University	
ChatGPT for Academic Researchers Program, Approved Participant	Aug. 2026
OpenAI	
Google Cloud Research Credits Program, Credit Recipient	Jun. 2026
Google Cloud	
Google Cloud Research Credits Program, Credit Recipient	Aug. 2025
Google Cloud	
2025 ITE Great Lakes District Student Paper Award & GLITE Endowment Fund	Jun. 2025
Great Lakes District of Institute of Transportation Engineers (GLITE)	
Honorable Mention, 2025 Clifford Spiegelman Student Paper Competition	Jan. 2025
The Transportation Statistics Interest Group of the American Statistical Association	
2025 KOTAA Travel Grant Award	Jan. 2025
Korean Transportation Association in America (KOTAA)	
Student Paper Award	Dec. 2024
ITE Indiana	
Kinnier Graduate Scholarship	Oct. 2024
Lyles School of Civil and Construction Engineering, Purdue University	

STV Civil Engineering Graduate Assistantship Endowment Lyles School of Civil and Construction Engineering, Purdue University	Oct. 2024
2024 KOTAA Travel Grant Award Korean Transportation Association in America (KOTAA)	Jan. 2024
Crooks Travel Scholarship Lyles School of Civil Engineering, Purdue University	Sep. 2023
Outstanding Paper Award 78th Conference of Korean Society of Transportation, Korean Society of Transportation	Mar. 2018
Outstanding Paper Award 74th Conference of Korean Society of Transportation, Korean Society of Transportation	Feb. 2016

Selected Presentations and Invited Talks

1. Physics-Informed AI for Transportation Systems. *CGN 6425: Applied Data Science in Civil and Environmental Engineering, University of Florida*. April 2, 2026.
2. Needs and Gap Assessment of INDOT Workforce Development. *2025 Purdue Road School Transportation Conference and Expo*. March 18, 2025.
3. A Physics-Informed Machine Learning for Generalized Bathtub Model in Large-Scale Urban Networks. *TRBACP50 Committee on Traffic Flow Theory and Characteristics General Webinar Series*. March 17, 2023.

Academic Service and Professional Activities

ACADEMIC SERVICE

- **Journal Referee:** IEEE Transactions on Intelligent Transportation Systems; Transportation Research Part A: Policy and Practice; Transportation Research Part B: Methodological; Transportation Research Part C: Emerging Technologies; Transportation Research Part F: Traffic Psychology and Behaviour; Transportation Research Interdisciplinary Perspectives; Transportation; Engineering Applications of Artificial Intelligence; Scientific Reports; Electric Power Systems Research; The Journal of Supercomputing; Journal of Cloud Computing; Transportation Research Record; Journal of the Transportation Research Board; Journal of Transportation Engineering, Part A: Systems; Data Science for Transportation
- **Conference Referee:** Transportation Research Board; IEEE Intelligent Transportation Systems Conference (ITSC); ISTTT25 (Co-reviewer); ISTTT26 (Co-reviewer)

PROFESSIONAL ACTIVITIES

- **Association for Computing Machinery (ACM):** Professional Member (Aug. 2025 - Current)
- **American Society of Civil Engineers (ASCE):** Associate Member (May 2025 - Current); Student Member (Sep. 2022 - Apr. 2025); **AI Committee:** Corresponding Member (Aug. 2025 - Current)
- **Institute of Electrical and Electronics Engineers (IEEE):** Member (May 2025 - Current)
- **Institute for Operations Research and the Management Sciences (INFORMS):** Member (May 2023 - Current)
- **Institute of Transportation Engineers (ITE) Purdue Student Chapter:** Communications Director (Aug. 2024 - July 2025); Member (Aug. 2020 - Current)

Technical Skills

- **Transportation Modeling and Simulation :** SUMO, PTV Vissim, TransModeler, TransCAD
- **Geospatial and Statistical Analysis :** ArcGIS, SAS, R, GeoPandas, SPSS, NLOGIT
- **Programming, Machine Learning, and Optimization :** Python, PyTorch, TensorFlow, Keras, PySpark, Java, MATLAB, CPLEX
- **Reproducible Research and Documentation :** Git, Jupyter, LaTeX